11.2 GMMs for Anomaly Detection

Low density under the mixture as an anomaly score.

These notes walk through the lecture slides. The original deck is unchanged; use **(slides)** on the course hub if you want that PDF.

Figures and notes follow Géron, *Hands-On Machine Learning*; Deisenroth, Faisal, and Ong, *Mathematics for Machine Learning*; and Theodoridis.

1. Anomaly detection

Find rows that do not look like the rest. Those are **anomalies** (outliers). The bulk are **inliers**.

- **Fraud.** Odd transactions or activity.
- **Manufacturing.** Defects, failing machines.
- **Preprocessing.** Drop junk before you train, so the model is less pushed around by noise.

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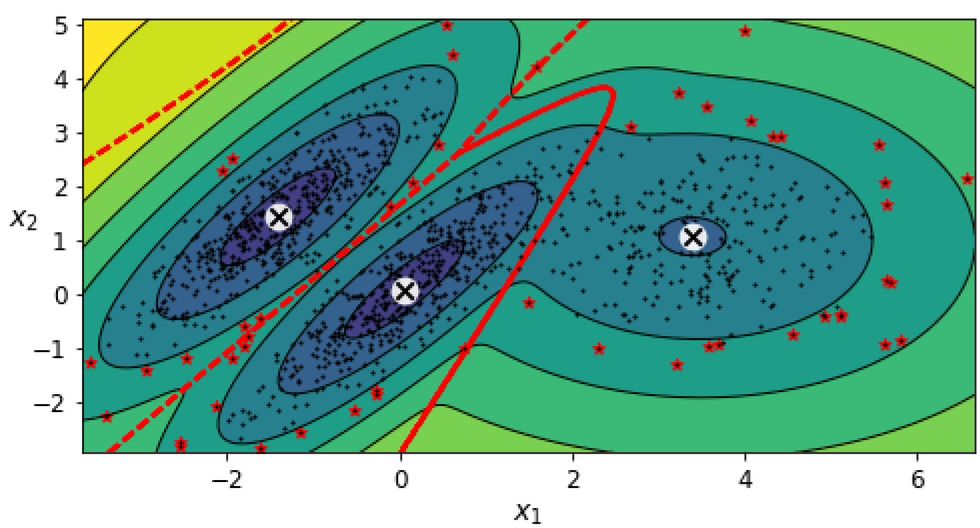
2. GMMs as the density

Normal behavior. Fit a mixture of Gaussians to ordinary data. Points that do not fit are anomalies.

Flexible. Network traffic, money, sensors — one tool, many shapes of "normal."

Density. Low-density regions are the anomalies. Set the threshold from domain knowledge, or from a known defect rate. Example: 4% defective in a factory → flag the 4% lowest-density points. Stars in the figure are the anomalies.



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3. Limits

Issue	What goes wrong
Gaussian assumption	If "normal" is not a mixture of ellipsoids, the density is a bad map.
Initialization	Bad starts → local optima, or a poor picture of normal.
Nonlinear structure	The model is a sum of Gaussians, not a nonlinear manifold.
High dimension	Many parameters; curse of dimensionality; overfitting or slowness.
Bias from outliers	The mixture tries to fit **everything**, including the junk. Too many outliers, and some of them start to look "normal."

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4. Tips

Features. Choose them. PCA if the ambient dimension is large.

Evaluation. Precision, recall, F1. Cross-validation or a holdout.

Hyperparameters. Number of components, covariance type, regularization.

Practice

1. How would you turn a GMM density into an anomaly score?
2. Why can too many outliers bias a GMM's picture of "normal"?
3. Name one reason a GMM can fail as an anomaly detector even when you set the density threshold carefully.